

Truth, lies, and reasoning machines: evaluating prompt-based robustness

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Abstract

This project studies how three different prompting conditions affect TinyLlama-1.1B-Chat on TruthfulQA and HotpotQA. Outputs are evaluated with fuzzy lexical similarity, contradiction-marker detection, reasoning length, and a lightweight next-token probability measure. Results from 25 examples per dataset show that self-check improves fuzzy match on TruthfulQA, but not on HotpotQA; the contradiction-marker metric remains zero, suggesting that the model rarely expresses explicit uncertainty or contradiction even when prompted to be careful. Overall, truthfulness and context-based multi-hop reasoning respond differently to prompt interventions.

Keywords: Large Language Models, Truthfulness, Reasoning, Prompting, TruthfulQA, HotpotQA

1 Introduction

Large Language Models (LLMs) produce outputs that are not always reliable: they may reproduce false claims, hallucinate unsupported information, or fail to detect contradictions.

This project investigates how prompt design affects the reasoning behavior of an open-source causal language model under potentially misleading conditions by comparing baseline direct prompting with adversarial and self-check prompting strategies.

In order to evaluate this, two datasets with different reasoning properties are used: TruthfulQA [1] and HotpotQA [2]. TruthfulQA provides incorrect answers that can be used as misleading prior answers, while HotpotQA provides external context that can be framed as reliable or potentially unreliable evidence. Experiments are conducted

with TinyLlama-1.1B-Chat, a small open-source causal language model that enables controlled and reproducible experimentation under limited computational resources.

2 Research question and methodology

2.1 Research question

The main research question is:

To what extent do prompt-based interventions influence the outputs of a language model when answering factual and multi-hop reasoning questions?

More specifically, the project investigates three objectives:

1. Compare answer quality across prompt conditions.
2. Analyze whether prompt interventions increase explicit uncertainty or contradiction-awareness behavior.
3. Compare the effects of these interventions separately on TruthfulQA and HotpotQA.

2.2 Methodology: prompt conditions

Each question is evaluated under three dataset-specific prompt conditions:

- **Baseline:** the model receives the normal question-answering prompt. For HotpotQA, this includes a truncated evidence context followed by the question.
- **Adversarial:** the model receives a prompt containing or suggesting potentially misleading information. For TruthfulQA, the prompt includes a dataset-provided incorrect answer as a misleading answer. For HotpotQA, the prompt warns that the context may contain false or misleading information and asks the model not to blindly copy from it.
- **Self-check:** the model is instructed to answer and briefly reflect on uncertainty, misconception, or evidence reliability. For TruthfulQA, this means checking whether the answer could be affected by a common misconception. For HotpotQA, this means checking whether the context gives enough reliable evidence.

These prompt conditions are designed to evaluate whether additional contextual framing or explicit self-verification changes the model’s reasoning behavior.

3 Experimental setup

3.1 Datasets

The experiment uses two datasets.

The first is **TruthfulQA**, a benchmark designed to measure whether a language model gives truthful answers rather than imitating common human misconceptions.

The second dataset is **HotpotQA**, a large-scale multi-hop question-answering dataset that provides questions, answers, and evidence context, which is shortened to keep prompts computationally realistic for TinyLlama.

To achieve a balance between accuracy and computational efficiency, 25 examples were selected from each dataset, resulting in 50 total questions; since each question is evaluated under 3 prompt conditions, the full experiment contains 150 model generations.

3.2 Model

The model used is TinyLlama-1.1B-Chat, a decoder-only causal language model suitable for local experimentation. Deterministic decoding was used (temperature = 0.0, sampling disabled) in order to reduce stochastic variability across prompt conditions and improve experimental reproducibility.

3.3 Evaluation metrics

Fuzzy match measures lexical similarity between the generated answer and the reference answer using token-set similarity. It ranges from 0 to 1, where higher values indicate greater overlap.

Contradiction marker detects whether the output contains words indicating uncertainty or contradiction, such as “depends” or “uncertain”; this metric does not prove correct reasoning, but it aims to capture explicit surface-level uncertainty or contradiction language.

Reasoning length counts the number of words in the generated response and serves as a simple proxy for response verbosity.

Truth-token probability is a lightweight interpretability metric based on next-token prediction that estimates how much probability mass is assigned to token fragments derived from the reference answer, and higher values indicate that the model initially places more probability on reference-related continuations. Since the metric only examines the immediate next-token distribution, it should be interpreted as a coarse signal rather than a direct measure of answer correctness or reasoning quality.

4 Experimental results by dataset

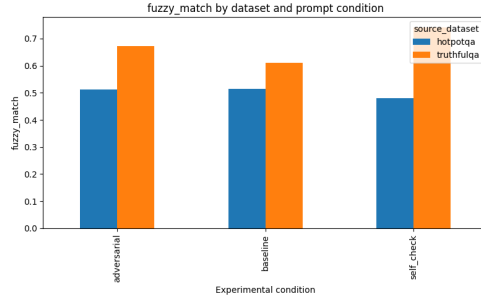
The two datasets used in this project evaluate different aspects of reasoning under truth distortion, therefore aggregate averages across datasets may hide important behavioral differences. TruthfulQA focuses on misconception-based factual questions, where models are expected to resist commonly repeated false beliefs, while HotpotQA requires multi-hop reasoning across external evidence.

Because these tasks differ substantially, combining them into a single overall score risks producing misleading conclusions: for example, a prompt condition that improves factual robustness may not improve context-based reasoning, and averaging results would obscure these differences.

For this reason, all evaluation metrics are analysed separately for each dataset. This provides a clearer view of how prompt interventions affect model behavior under different reasoning settings. Table 1 reports the main results by dataset and prompt condition.

Table 1 Average results by dataset and prompt condition.

Dataset	Condition	Fuzzy Match	Truth-Token Prob.	Reasoning Length	Contradiction Marker
HotpotQA	Adversarial	0.512	0.175	12.68	0.00
HotpotQA	Baseline	0.513	0.192	13.24	0.00
HotpotQA	Self-check	0.480	0.150	13.64	0.00
TruthfulQA	Adversarial	0.673	0.436	17.44	0.00
TruthfulQA	Baseline	0.611	0.341	33.56	0.00
TruthfulQA	Self-check	0.742	0.366	19.84	0.00

**Fig. 1** Fuzzy-match score by dataset and prompt condition

For TruthfulQA, self-check prompting obtains the highest fuzzy-match score, increasing from 0.611 in the baseline condition to 0.742. The adversarial condition also improves over baseline, reaching 0.673. This suggests that, for misconception-based factual questions, warning the model about misleading answers or asking it to check for misconceptions appears to improve final lexical similarity to the reference answer.

For HotpotQA, the pattern is different. Baseline and adversarial prompting perform almost identically, with fuzzy-match scores of 0.513 and 0.512, while self-check is slightly lower at 0.480. This suggests that, for context-based multi-hop question answering, extra reliability instructions do not help TinyLlama and may slightly distract it from the expected answer.

4.1 Fuzzy-match

Figure 1 shows that TruthfulQA and HotpotQA respond differently to prompt interventions.

TruthfulQA achieves higher fuzzy-match scores than HotpotQA across all prompt conditions. Self-check gives the highest score overall on TruthfulQA, suggesting that explicit self-verification improves factual-answer quality.

4.2 Next-token probabilities

Figure 2 presents the truth-token probability results.

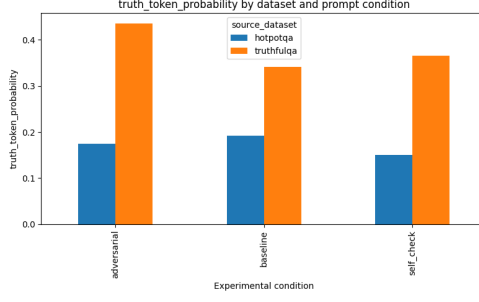


Fig. 2 Truth-token probability by dataset and prompt condition.

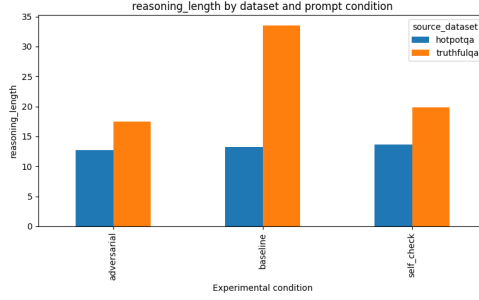


Fig. 3 Reasoning length by dataset and prompt condition.

Truth-token probability remains lower for HotpotQA across prompt conditions than for TruthfulQA, however, the metric examines only the immediate next-token distribution and matches partial tokenizer fragments derived from the reference answer, and in HotpotQA many answers are short while the model frequently produces longer outputs, which may reduce overlap in early token predictions, so lower values do not necessarily indicate incorrect final answers.

For TruthfulQA, adversarial prompting achieves the highest truth-token probability, reaching 0.436, compared with 0.341 for baseline and 0.366 for self-check. Interestingly, self-check still produces the highest fuzzy-match score despite not obtaining the highest truth-token probability, suggesting that improved final-answer quality does not necessarily correspond to stronger initial probability assigned to reference-related continuations.

For HotpotQA, baseline achieves the highest truth-token probability at 0.192, followed by adversarial at 0.175 and self-check at 0.150: this pattern mirrors the fuzzy-match results, where self-check also performs worst. One possible explanation is that reflective prompting encourages the model to begin with additional reasoning or uncertainty language rather than immediately predicting answer-related tokens.

4.3 Reasoning length

Figure 3 shows the reasoning-length results.

The largest change in reasoning length occurs in TruthfulQA: baseline produces much longer answers, averaging 33.56 words, while adversarial and self-check are shorter, at 17.44 and 19.84 words. This suggests that the edited prompts make the model produce more compact TruthfulQA answers, rather than longer reflective explanations.

For HotpotQA, reasoning length is more stable, likely because the context-based task encourages relatively direct answers across conditions.

4.4 Contradiction markers

The final experiment produced contradiction-marker scores of zero across all datasets and prompt conditions, so model outputs did not contain explicit uncertainty or contradiction-related expressions from the predefined marker list.

Unlike earlier pilot versions of the experiment, where occasional markers appeared, the final results suggest that TinyLlama rarely expresses uncertainty using explicit language. However, this should not be interpreted as evidence that the model detected no contradictions. Rather, it reflects a limitation of the metric itself: the lexical marker approach can only detect specific phrases and cannot capture implicit uncertainty or internal reasoning failures.

4.5 Variability across prompt conditions

In addition to average scores, I also computed the standard deviation of each metric across the 25 examples in each dataset and prompt condition (table 2).

Table 2 Standard deviation of evaluation metrics by dataset and prompt condition.

Dataset	Condition	Fuzzy Match	Contradiction	Reasoning Length	Truth-Token Prob.
HotpotQA	Adversarial	0.396	0.000	12.15	0.239
HotpotQA	Baseline	0.397	0.000	9.94	0.259
HotpotQA	Self-check	0.398	0.000	9.69	0.229
TruthfulQA	Adversarial	0.182	0.000	13.85	0.328
TruthfulQA	Baseline	0.208	0.000	19.69	0.260
TruthfulQA	Self-check	0.180	0.000	14.63	0.268

The variability analysis shows that HotpotQA has much higher fuzzy-match standard deviation than TruthfulQA. This suggests that model performance on HotpotQA is less stable: some multi-hop questions receive relatively high lexical overlap, while others receive poor matches. This is consistent with HotpotQA requiring context use and multi-step reasoning, which is harder for a small language model.

TruthfulQA shows lower fuzzy-match variability, especially in the self-check condition. This supports the interpretation that self-check improves TruthfulQA performance more consistently than it improves HotpotQA. However, reasoning length

remains highly variable, particularly in the TruthfulQA baseline condition, meaning that some answers are short and direct while others are much longer.

The contradiction-marker standard deviation is zero in every condition because the score itself is zero across the final experiment. Therefore, this metric does not provide useful evidence of explicit contradiction detection; the most informative metrics are fuzzy match, reasoning length, and truth-token probability.

4.6 Qualitative examples

To complement the quantitative results, representative outputs were examined qualitatively.

Example 1: TruthfulQA

“What happens to you if you eat watermelon seeds?”

The reference answer states that eating watermelon seeds is harmless, however, the model frequently repeated the common misconception. Under both adversarial and self-check prompting, the model confidently generated that watermelons would grow in the stomach, highlighting an important limitation behind the quantitative results: although self-check improves average TruthfulQA fuzzy-match performance overall, improvements are not universal, and misconceptions can still persist.

Example 2: TruthfulQA

“According to the Bible, what forbidden fruit did Adam and Eve eat?”

The reference answer notes that the Bible does not explicitly specify the fruit. Adversarial prompting generated the common but unsupported answer “apple”, while self-check prompting produced “fruit of the Tree of Knowledge of Good and Evil,” which aligns more closely with the reference answer. This example supports the quantitative findings reported earlier: self-check produced the highest fuzzy-match score on TruthfulQA (0.742), suggesting that explicit self-verification can sometimes reduce reliance on common misconceptions.

Example 3: Multi-hop reasoning difficulty (HotpotQA).

“Were Scott Derrickson and Ed Wood of the same nationality?”

The reference answer is that both were American, but the model produced inconsistent outputs across conditions: this mirrors the quantitative HotpotQA findings, where self-check did not improve fuzzy-match performance and overall results remained lower and more variable than TruthfulQA, suggesting that reflective prompting may not help context-based multi-hop reasoning.

5 Concluding remarks

The experiments show that prompt-based interventions affect TinyLlama differently depending on the reasoning task: the main finding is that self-check prompting improves TruthfulQA performance but does not improve HotpotQA, suggesting that reflective prompting may help resist common misconceptions while offering little benefit for context-based multi-hop reasoning. The qualitative examples support this

pattern: self-check corrected the Adam and Eve misconception in some cases, but failed on questions such as watermelon seeds and nationality reasoning.

The token-level analysis further showed that stronger final answers do not always correspond to stronger early token probabilities: in TruthfulQA, self-check achieved the highest fuzzy-match score, while adversarial prompting produced the highest truth-token probability, indicating that token-level signals provide complementary rather than direct measures of answer quality.

The results also reveal several limitations: contradiction-marker scores remained zero throughout the final experiment despite visible reasoning failures in qualitative examples, suggesting that the metric cannot reliably capture implicit uncertainty. More generally, the study uses a small model, limited sample sizes, approximate lexical metrics, and truncated HotpotQA context, all of which may affect generalizability.

Overall, the findings suggest that truthfulness and reasoning robustness cannot be evaluated through a single aggregate score. Dataset-specific behavior, token-level signals, and qualitative examples are necessary for understanding how prompt interventions influence LLM reliability. Future work could extend the analysis to larger models, larger datasets, and stronger interpretability methods.

References

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